4.2.3 Organic synthesis

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
Practical skills		
(a)	the techniques and procedures for: **(i)** use of Quickfit apparatus including for distillation and heating under reflux (ii) preparation and purification of an organic liquid including: • use of a separating funnel to remove an organic layer from an aqueous layer • drying with an anhydrous salt (e.g. $MgSO_4$, $CaCl_2$) • redistillation	**PAG5** HSW4 Opportunities to carry out experimental and investigative work.
Synthetic routes		
(b)	for an organic molecule containing several functional groups: **(i)** identification of individual functional groups (ii) prediction of properties and reactions	Learners will be expected to identify the functional groups encountered in Module 4. HSW3 Development of synthetic routes.
(c)	two-stage synthetic routes for preparing organic compounds.	Learners will be expected to be able to devise two-stage synthetic routes by applying transformations between all functional groups encountered up to this point of the specification. Extra information may be provided on exam papers to extend the learner's toolkit of organic reactions. HSW3 Development of synthetic routes.